Additional file 4. Supplementary tables added at peer review (Tables S2 to S8)

Blood metal and nutritional biomarkers did not improve identification of low cognitive performance beyond demographics in older US adults: a pre-registered analysis of NHANES 2011-2014. Kancharla S, Patel N.

Every analysis in this file was requested at peer review and is post hoc unless marked as pre-specified. All were recorded in a dated amendment to the pre-registration (Additional file 1, Amendment 1, 2026-11-04) before they were run. Tables S2 to S5 are descriptive and involve no predictor-outcome relationship. Tables S6 and S8 report ΔAUC with paired-bootstrap intervals from 2,000 resamples of participants; contrasts reported in the original submission reproduce their original resamples exactly, and each new contrast uses its own fixed-seed set of 2,000 resamples. Every number is produced by the scripts `src/11_revision_models.py` to `src/14_revision_tables.py` in the public repository.

Table S2: Missing cognitive scores by test and by reason, all 3,472 examined participants aged 60 and over. Reasons are the NHANES codes `CFASTAT` (proxy respondent, language not offered, no test done) and the test-specific reason codes `CFDCRNC` (CERAD), `CFDARNC` (animal fluency) and `CFDDRNC` (DSST), checked against the CDC codebooks for CFQ_G and CFQ_H. A CERAD reason applies to both CERAD scores.

Reason	CERAD immediate	CERAD delayed	Animal fluency	DSST
Not eligible: proxy respondent	49	49	49	49
Not eligible: language not offered	25	25	25	25
No cognitive test done	202	202	202	202
Did not consent to audio recording	13	13	13	0
Communication problem	6	6	3	11
Physical limitation	4	4	3	22
Quit or gave up	10	15	0	0
Refused	10	10	10	34
Equipment failure	22	22	39	0
Other	0	0	0	2
Missing, no reason recorded	0	0	18	113
Total missing	341	346	362	458
Score present	3131	3126	3110	3014

Table S3: Percentage of participants missing each cognitive score, and with fewer than three scores (and therefore no outcome label), by demographic group, among the 3,472 examined participants aged 60 and over.

Group	*n*	CERAD immediate	CERAD delayed	Animal fluency	DSST	Fewer than 3 scores
All examined, aged 60+	3472	9.8%	10.0%	10.4%	13.2%	10.0%
Age 60 to 69	1786	7.7%	7.8%	8.1%	8.8%	7.7%
Age 70 to 79	1028	8.9%	9.1%	10.2%	13.0%	9.2%
Age 80 (top-coded, 80 and over)	658	17.0%	17.2%	17.2%	25.2%	17.5%
Male	1687	10.0%	10.1%	10.8%	12.9%	10.2%
Female	1785	9.7%	9.8%	10.1%	13.4%	9.9%
Mexican American	325	11.4%	11.4%	12.6%	19.1%	11.7%
Other Hispanic	346	7.5%	8.4%	8.4%	11.6%	8.4%
Non-Hispanic White	1580	8.0%	8.1%	8.1%	10.1%	8.1%
Non-Hispanic Black	837	8.8%	8.6%	9.6%	13.6%	8.7%
Non-Hispanic Asian	331	21.5%	22.1%	23.0%	21.8%	22.1%
Other/multiracial	53	13.2%	13.2%	15.1%	20.8%	13.2%
Less than 9th grade	511	21.5%	22.1%	23.3%	32.1%	22.5%
9-11th grade	507	10.8%	11.0%	11.4%	15.8%	11.2%
High school grad/GED	797	9.4%	9.7%	9.8%	11.9%	9.5%

Group	*n*	CERAD immediate	CERAD delayed	Animal fluency	DSST	Fewer than 3 scores
Some college/AA degree	911	5.5%	5.5%	6.4%	7.5%	5.6%
College graduate or above	739	6.5%	6.4%	6.2%	6.4%	6.2%
Education missing	7	42.9%	42.9%	42.9%	57.1%	42.9%

Table S4: Participants with three against four component cognitive scores, in the labelled sample ($n = 3{,}124$) and in the analytic sample ($n = 1{,}784$). A participant with three scores has a composite built from different content. Values are *n* (%) unless stated.

	Labelled sample		Analytic sample	
Characteristic	Four scores	Three scores	Four scores	Three scores
n	2934	190	1694	90
Age, years, mean (SD)	69.5 (6.8)	72.8 (7.0)	69.2 (6.7)	72.0 (7.1)
Female	1506 (51.3%)	103 (54.2%)	859 (50.7%)	45 (50.0%)
Less than high school education	746 (25.4%)	100 (52.6%)	396 (23.4%)	47 (52.2%)
Hispanic (Mexican American or other Hispanic)	553 (18.8%)	51 (26.8%)	301 (17.8%)	21 (23.3%)
Non-Hispanic Black	698 (23.8%)	66 (34.7%)	397 (23.4%)	28 (31.1%)
Income-to-poverty ratio, mean (SD)	2.60 (1.60)	1.93 (1.42)	2.62 (1.61)	2.16 (1.49)
Tested in Spanish or an Asian language	346 (11.8%)	45 (23.7%)	187 (11.0%)	22 (24.4%)
Missing score: CERAD immediate recall	0 (0.0%)	3 (1.6%)	0 (0.0%)	3 (3.3%)
Missing score: CERAD delayed recall	0 (0.0%)	3 (1.6%)	0 (0.0%)	3 (3.3%)
Missing score: animal fluency	0 (0.0%)	35 (18.4%)	0 (0.0%)	20 (22.2%)
Missing score: DSST	0 (0.0%)	149 (78.4%)	0 (0.0%)	64 (71.1%)
Composite z, mean (SD)	0.05 (0.76)	−0.84 (0.89)	0.05 (0.76)	−0.85 (0.90)
Low cognitive performance	666 (22.7%)	115 (60.5%)	373 (22.0%)	58 (64.4%)

Table S5: Participants with an outcome label (three or four scores) against those without one (zero to two scores), among the 3,472 examined participants aged 60 and over. Values are *n* (%) unless stated.

Characteristic	Labelled	Unlabelled
n	3124	348
Age, years, mean (SD)	69.7 (6.8)	72.3 (7.4)
Age 80 (top-coded)	543 (17.4%)	115 (33.0%)
Female	1609 (51.5%)	176 (50.6%)
Less than high school education	846 (27.1%)	172 (49.4%)
Hispanic (Mexican American or other Hispanic)	604 (19.3%)	67 (19.3%)
Non-Hispanic Black	764 (24.5%)	73 (21.0%)
Income-to-poverty ratio, mean (SD)	2.56 (1.60)	1.86 (1.34)
Ever told had a stroke	237 (7.6%)	62 (17.8%)
Ever told had diabetes	892 (28.6%)	103 (29.6%)
PHQ-9, median [IQR]	2.00 [0.00, 4.00]	1.00 [0.00, 5.00]
Proxy respondent or no test done	0 (0.0%)	251 (72.1%)
Not tested: language not offered	0 (0.0%)	25 (7.2%)

Table S6: Every pairwise ΔAUC contrast among the six models under each algorithm, and each nonlinear learner against logistic regression within each model, on the locked analytic sample ($n = 1{,}784$, 431 events). Contrasts are oriented so that the larger or later model comes first. "Nested" marks contrasts in which the second model's features are a subset of the first's. The final column applies the 0.02 criterion symmetrically: "yes" would mean the 95% interval lies wholly beyond ±0.02; no contrast meets it. Contrasts involving M4 and all contrasts not reported in the original submission are post hoc.

Contrast	Nested	ΔAUC	95% CI	90% CI	Beyond ±0.02
Logistic					
M1 − M0	yes	−0.0015	[−0.0052, 0.0023]	[−0.0047, 0.0018]	no
M2 − M0	yes	0.0011	[−0.0067, 0.0100]	[−0.0055, 0.0087]	no
M3 − M0	yes	−0.0019	[−0.0108, 0.0071]	[−0.0094, 0.0056]	no
M0+ − M0	yes	0.0139	[0.0026, 0.0255]	[0.0043, 0.0232]	no
M4 − M0	yes	0.0102	[−0.0030, 0.0226]	[−0.0012, 0.0208]	no
M2 − M1	no	0.0027	[−0.0061, 0.0116]	[−0.0046, 0.0099]	no
M3 − M1	yes	−0.0003	[−0.0086, 0.0076]	[−0.0068, 0.0063]	no
M0+ − M1	no	0.0155	[0.0041, 0.0267]	[0.0058, 0.0251]	no
M4 − M1	yes	0.0118	[−0.0008, 0.0247]	[0.0012, 0.0227]	no
M3 − M2	yes	−0.0030	[−0.0058, −0.0002]	[−0.0053, −0.0006]	no
M0+ − M2	no	0.0128	[−0.0001, 0.0257]	[0.0017, 0.0237]	no
M4 − M2	yes	0.0091	[−0.0013, 0.0200]	[0.0004, 0.0183]	no
M0+ − M3	no	0.0158	[0.0033, 0.0295]	[0.0055, 0.0270]	no
M4 − M3	yes	0.0121	[0.0026, 0.0223]	[0.0037, 0.0209]	no
M4 − M0+	yes	−0.0037	[−0.0112, 0.0029]	[−0.0100, 0.0019]	no
Random forest					
M1 − M0	yes	−0.0058	[−0.0144, 0.0022]	[−0.0131, 0.0007]	no
M2 − M0	yes	−0.0028	[−0.0155, 0.0103]	[−0.0131, 0.0083]	no
M3 − M0	yes	−0.0038	[−0.0167, 0.0095]	[−0.0145, 0.0073]	no
M0+ − M0	yes	0.0160	[0.0056, 0.0266]	[0.0073, 0.0249]	no
M4 − M0	yes	0.0067	[−0.0077, 0.0212]	[−0.0053, 0.0182]	no
M2 − M1	no	0.0030	[−0.0106, 0.0154]	[−0.0085, 0.0139]	no
M3 − M1	yes	0.0021	[−0.0088, 0.0129]	[−0.0074, 0.0110]	no
M0+ − M1	no	0.0218	[0.0104, 0.0334]	[0.0122, 0.0314]	no
M4 − M1	yes	0.0125	[0.0003, 0.0251]	[0.0019, 0.0229]	no
M3 − M2	yes	−0.0010	[−0.0056, 0.0038]	[−0.0049, 0.0029]	no
M0+ − M2	no	0.0188	[0.0051, 0.0322]	[0.0072, 0.0304]	no
M4 − M2	yes	0.0095	[0.0016, 0.0170]	[0.0030, 0.0159]	no
M0+ − M3	no	0.0197	[0.0055, 0.0338]	[0.0077, 0.0315]	no
M4 − M3	yes	0.0104	[0.0037, 0.0173]	[0.0051, 0.0160]	no
M4 − M0+	yes	−0.0093	[−0.0196, 0.0013]	[−0.0181, −0.0002]	no
Gradient boosting					
M1 − M0	yes	−0.0130	[−0.0212, −0.0046]	[−0.0198, −0.0060]	no
M2 − M0	yes	−0.0017	[−0.0143, 0.0113]	[−0.0125, 0.0094]	no
M3 − M0	yes	−0.0066	[−0.0199, 0.0068]	[−0.0179, 0.0048]	no
M0+ − M0	yes	0.0164	[0.0051, 0.0288]	[0.0070, 0.0267]	no

Contrast	Nested	ΔAUC	95% CI	90% CI	Beyond ±0.02
M4 − M0	yes	0.0071	[−0.0074, 0.0215]	[−0.0054, 0.0196]	no
M2 − M1	no	0.0113	[−0.0019, 0.0240]	[−0.0001, 0.0217]	no
M3 − M1	yes	0.0064	[−0.0052, 0.0173]	[−0.0033, 0.0155]	no
M0+ − M1	no	0.0294	[0.0163, 0.0429]	[0.0182, 0.0405]	no
M4 − M1	yes	0.0201	[0.0061, 0.0345]	[0.0087, 0.0320]	no
M3 − M2	yes	−0.0049	[−0.0094, −0.0003]	[−0.0087, −0.0010]	no
M0+ − M2	no	0.0181	[0.0021, 0.0325]	[0.0051, 0.0303]	no
M4 − M2	yes	0.0088	[−0.0006, 0.0183]	[0.0010, 0.0171]	no
M0+ − M3	no	0.0230	[0.0063, 0.0385]	[0.0096, 0.0359]	no
M4 − M3	yes	0.0137	[0.0051, 0.0227]	[0.0064, 0.0214]	no
M4 − M0+	yes	−0.0093	[−0.0201, 0.0018]	[−0.0186, −0.0001]	no
Random forest vs logistic					
M0: random forest − logistic	no	−0.0040	[−0.0122, 0.0052]	[−0.0112, 0.0038]	no
M1: random forest − logistic	no	−0.0083	[−0.0185, 0.0017]	[−0.0168, 0.0002]	no
M2: random forest − logistic	no	−0.0079	[−0.0197, 0.0030]	[−0.0180, 0.0015]	no
M3: random forest − logistic	no	−0.0059	[−0.0175, 0.0060]	[−0.0158, 0.0038]	no
M0+: random forest − logistic	no	−0.0020	[−0.0127, 0.0079]	[−0.0114, 0.0064]	no
M4: random forest − logistic	no	−0.0076	[−0.0194, 0.0048]	[−0.0179, 0.0027]	no
Gradient boosting vs logistic					
M0: gradient boosting − logistic	no	−0.0039	[−0.0126, 0.0051]	[−0.0113, 0.0034]	no
M1: gradient boosting − logistic	no	−0.0153	[−0.0246, −0.0061]	[−0.0232, −0.0078]	no
M2: gradient boosting − logistic	no	−0.0067	[−0.0161, 0.0033]	[−0.0149, 0.0016]	no
M3: gradient boosting − logistic	no	−0.0086	[−0.0187, 0.0007]	[−0.0171, −0.0004]	no
M0+: gradient boosting − logistic	no	−0.0014	[−0.0106, 0.0076]	[−0.0091, 0.0061]	no
M4: gradient boosting − logistic	no	−0.0070	[−0.0176, 0.0030]	[−0.0160, 0.0016]	no

Table S7: Out-of-fold discrimination and calibration for every model and algorithm, including M4 (post hoc), on the locked analytic sample. Calibration intercept is calibration-in-the-large from the offset model.

Model	Algorithm	AUC	AUPRC	Brier	Calibration slope	Calibration intercept
M0	Logistic	0.7884	0.5471	0.1459	1.088	0.0012
M0	Random forest	0.7843	0.5363	0.1466	1.148	0.0037
M0	Gradient boosting	0.7845	0.5399	0.1470	0.898	0.0041
M1	Logistic	0.7868	0.5456	0.1465	1.069	0.0015
M1	Random forest	0.7785	0.5290	0.1491	1.193	−0.0040
M1	Gradient boosting	0.7715	0.5212	0.1506	0.883	0.0087
M2	Logistic	0.7895	0.5420	0.1461	1.026	0.0021
M2	Random forest	0.7815	0.5471	0.1488	1.321	−0.0110
M2	Gradient boosting	0.7828	0.5209	0.1488	0.912	0.0201
M3	Logistic	0.7865	0.5392	0.1470	1.004	0.0015
M3	Random forest	0.7806	0.5384	0.1501	1.410	−0.0118
M3	Gradient boosting	0.7779	0.5201	0.1497	0.904	0.0241
M0+	Logistic	0.8023	0.5618	0.1423	1.042	0.0027
M0+	Random forest	0.8003	0.5525	0.1448	1.383	0.0037
M0+	Gradient boosting	0.8009	0.5428	0.1438	0.914	0.0098
M4	Logistic	0.7986	0.5537	0.1441	0.984	0.0017
M4	Random forest	0.7910	0.5477	0.1486	1.527	−0.0118
M4	Gradient boosting	0.7916	0.5305	0.1466	0.933	0.0270

Table S8: Margin analysis for every pre-specified and post hoc sensitivity analysis. "Excluded above" is the upper bound of the 90% interval: improvements larger than this are excluded at the one-sided 5% level. "TOST margin" is the narrowest symmetric margin δ for which equivalence within $\pm\delta$ is shown by two one-sided tests at $\alpha = 0.05$, i.e. the larger of the absolute 90% bounds. Pre-specified sensitivity analyses used L2-penalized logistic regression with a fixed penalty ($C = 1$), as in the original submission; the primary analysis, M4 and the post hoc analyses used the tuned primary pipeline. For the demographically adjusted outcome, AUC was 0.56 to 0.62 for every model, so its increments are on a different baseline from the others.

Analysis	Contrast	n	Events	ΔAUC	95% CI	90% CI	Excluded above	TOST margin
Primary analysis (pre-specified)	M3 − M0	1784	431	−0.0019	[−0.0108, 0.0071]	[−0.0094, 0.0056]	0.0056	0.0094
Biomarkers added to the clinical baseline (post hoc)	M4 − M0+	1784	431	−0.0037	[−0.0112, 0.0029]	[−0.0100, 0.0019]	0.0019	0.0100
Primary specification, phase 6 pipeline (pre-specified)	M3 − M0	1784	431	−0.0020	[−0.0103, 0.0066]	[−0.0090, 0.0055]	0.0055	0.0090
20th percentile cutoff (pre-specified)	M3 − M0	1784	337	−0.0019	[−0.0130, 0.0092]	[−0.0109, 0.0076]	0.0076	0.0109
10th percentile cutoff (pre-specified)	M3 − M0	1784	164	−0.0023	[−0.0200, 0.0153]	[−0.0170, 0.0132]	0.0132	0.0170
DSST not completed coded low (pre-specified)	M3 − M0	1784	441	−0.0020	[−0.0105, 0.0063]	[−0.0093, 0.0049]	0.0049	0.0093
English-language administration only (pre-specified)	M3 − M0	1575	345	−0.0066	[−0.0159, 0.0028]	[−0.0144, 0.0015]	0.0015	0.0144
Multiple imputation, full labelled sample (pre-specified)	M3 − M0	3124	781	0.0015	[−0.0030, 0.0063]	[−0.0024, 0.0057]	0.0057	0.0057
Complete cognitive battery (4 of 4 scores) (post hoc)	M3 − M0	1694	373	−0.0045	[−0.0144, 0.0054]	[−0.0127, 0.0038]	0.0038	0.0127
Complete cognitive battery (4 of 4 scores) (post hoc)	M4 − M0+	1694	373	−0.0062	[−0.0140, 0.0019]	[−0.0127, 0.0005]	0.0005	0.0127
Demographically adjusted outcome (internal norms) (post hoc)	M3 − M0	1784	450	0.0362	[0.0068, 0.0667]	[0.0109, 0.0615]	0.0615	0.0615
Demographically adjusted outcome (internal norms) (post hoc)	M4 − M0+	1784	450	0.0093	[−0.0109, 0.0284]	[−0.0075, 0.0255]	0.0255	0.0255